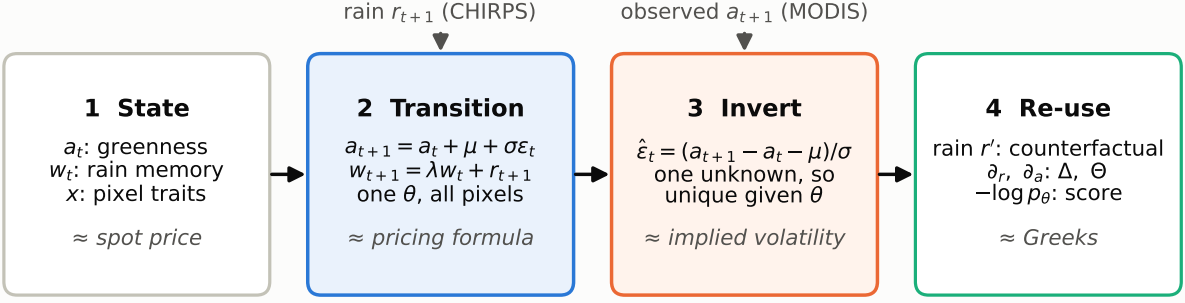
rain r_{t+1} (CHIRPS)
observed a_{t+1} (MODIS)
1 State
a_t: greenness
w_t: rain memory
x: pixel traits
≈ spot price
2 Transition
$a_{t+1} = a_t + \mu + \sigma\varepsilon_t$
$w_{t+1} = \lambda w_t + r_{t+1}$
one θ, all pixels
≈ pricing formula
3 Invert
$\hat{\varepsilon}_t = (a_{t+1} - a_t - \mu)/\sigma$
one unknown, so
unique given θ
≈ implied volatility
4 Re-use
rain r': counterfactual
∂_r, ∂_a: Δ, Θ
$-\log p_\theta$: score
≈ Greeks